

Clause combining in Tibeto-Burman

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1. Introduction

In this contribution, we focus on the clause combining strategies in Tibeto-Burman that have received most prominent attention in the literature: sentential nominalization, converbal clauses, and verb compounding, including serialization. Our main goal is to take stock of what is known about the structural and semantic properties of these strategies and provide a bird's eye view on their variation within Tibeto-Burman. Keeping with scope of the present volume, we refrain from integrating these structures into a universal typology. Instead, we use this review to develop and propose a questionnaire that serves as a basis for such a typology, inspired by previous work (Cristofaro 2003; Bickel 2010; van Gijn et al. 2015; van Gijn 2016). The goal of the questionnaire is to enable future research into the fine-grained variation in Tibeto-Burman clause combining strategies, as these are often characterized by functional and structural overlap.

The contribution is organized as follows. We first describe sentential nominalization patterns in Tibeto-Burman, and especially the properties of what has been named the Standard Sino-Tibetan nominalization (§2), with a focus on the functional convergence of Tibeto-Burman nominalized structures. We then describe converbal structures, and the properties of the South Asian converb (§3). Section 4 is dedicated to verb compounding and serialization structures. In Section 5, we introduce a typological questionnaire which aims at capturing clause combining strategies in Tibeto-Burman, and we describe the rationale and the design principles. We conclude our contribution with a short summary (§6). The questionnaire is illustrated in the Appendix.

2. Standard Sino-Tibetan Nominalization and functional/structural convergence

A prominent feature of Tibeto-Burman nominalization is its polyfunctionality and versatility. Tibeto-Burman nominalized clauses cover several syntactic functions and structures, including nominalization, relativization, genitivization, and non-subordinated independent clauses. This functional syncretism has been referred to as the “Standard Sino-Tibetan Nominalization” pattern (Bickel 1999) and, due to its pervasiveness,¹ has received substantial attention in the literature (a non-exhaustive list of language-specific

¹ This morphological convergence of syntactic functions, especially of relativization and nominalization, is also found outside Tibeto-Burman and in broader Asia, including what has been referred to as the Indo-Altaic area (Masica 1976), and is attested, for example, in Yukaghir (Maslova 1999), Burushaski (Anderson 2002), and Mongolian (Binnick 1979).

and group-specific studies includes: Bickel 1995, 1999; Chalise 2005; DeLancey 1986, 1990, 1999, 2005; Ebert 1994; Genetti 1992; Genetti et al. 2008; Herring 1991; Konow 1908, 1911; Kölver 1977; Lahaussais 2002, 2003; LaPolla 2006; Matisoff 1972; Mazaudon 1978; Noonan 1997, 2005; O'Rourke 2000; Post 2011; Regmi 2005; Vittrant 2002; Watters 2008.

This section explores the syntactic and functional variation displayed by nominalized structures in encoding (verbal) complement clauses, converbal clauses, relative clauses, and non-embedded, independent clauses. The use of nominalized structures to encode these syntactic environments is widely attested across the Tibeto-Burman family. However, not all Tibeto-Burman languages syncretize these functions (more below). At the same time, other languages, such as Bodic and Tamangic (i.e., Chantyal, Noonan 1997; 2005) exhibit an even broader range of functions encoded through nominalization.

Clausal nominalization and complementation

Verbal complement clauses in several Tibeto-Burman languages show structural congruence with clausal nominalization, leading to significant variation in their structures (for a comparative survey across a range of Tibeto-Burman languages, see Genetti et al. 2008, on which this contribution heavily relies on; Genetti 1992, 2011). Variables include the semantic range of predicates selecting nominalized complements, co-reference constraints, and the argument status of the complement clause. In addition, Tibeto-Burman nominalized complements can be internally complex: they may be combined with other clauses according to the clause linking devices available in a language, allowing for complex structures such as double complementation, found in Mongsen Ao (Genetti et al. 2008: 121, from Coupe 2007a:238), embedded relative clauses and converbal clauses (Genetti et al. 2008: 120), as well as clause chains embedded as a single unit as an object complement. This is illustrated for example in Lahu² (Matisoff 2017: 928):

	<i>ṇàhi</i>	<i>ga</i>	<i>yù</i>	<i>qay</i>	<i>cí</i>	<i>cô</i>	<i>ve</i>	<i>lâ</i>
(1)	1PL	help	take	go	send	correct	NMLZ	PUF

'Should (we) make (them) help to take (it) away?'

Languages vary in the types of verbs selecting nominalized complements. Some languages lack a specific class of complement-taking verbs (e.g., Mongsen Ao, Genetti

² The examples and the respective glosses and English translations are extracted *verbatim* from the sources, with some minimal adjustments. The examples have been generated with pandoc-glossify (<https://github.com/tzakharko/pandoc-glossify>).

et al. 2008), while others restrict the selection to subsets of verbs with specific semantics, such as perception verbs like 'see', 'watch', and 'hear' (cf. Dongwang Tibetan, Dolakha Newar, Zhuokeji rGyalrong; see discussion and examples in Genetti et al. 2008), and verbs like 'want' (i.e., Manange, Genetti et al. 2008), 'be/not be the case' (e.g., in Dolakha Newar), or 'say' (again in Dolakha Newar). This aspect directly relates to the argument status of the nominalized complement clause, with some nominalized complements functioning as S arguments (e.g., with verbs like 'to be the case') or O arguments, particularly with perception verbs.

Another parameter of variation in nominalized complements is the obligatory or optional co-reference between the arguments of the matrix clause and those of the nominalized complement. Co-reference can be obligatory, as in Manange desiderative constructions (Genetti et al. 2008:119), while certain languages, such as Dolakha Newar (Genetti 2007: 409), allow raising (where *amta* 'her' is raised to the object function of *khonai* 'sees'):

	<i>cilā=n</i>	<i>ninpatti</i>	<i>āmta</i>		<i>kho-en</i>	<i>coŋ-gu</i>
(2)	goat=ERG	daily	3SDAT		cry-PART	stay-NOM1
	<i>khon-ai</i>					
	see-3SPR					

'The goat sees her crying every day.'

Languages may express complementation through a variety of complex structures. Caodeng rGyalrong complement clauses include finite (but inflectionally reduced, cf. Sun 2012) nominalized structures, non-nominalized "S-like" clauses ("sentence like", similarly to the "sentential complements" in typological literature, cf. Givón 1980, Noonan 1985), and irrealis and infinitival clauses (for distributional constraints, properties, and internal variation see Sun 2012). Dolakha Newar, in addition to nominalized clauses, has other complementation types, including finite and infinitival structures (Genetti 2007: 415-420). Japhug rGyalrong uses an extensive array of structures in addition to clausal nominalization for complementation, including typologically unusual constructions with fine-grained semantic distinctions (Jacques 2016). A type of complement clause in Karbi is expressed via the juxtaposition of the main verb and complement verb (Konnerth 2011).

Clausal nominalization and converbal clauses

Nominalized clauses can additionally function as converbal clauses in Tibeto-Burman, often converging with relativization (discussed below). The 'converbal use' of nominalization varies across Tibeto-Burman languages in terms of formal encoding

(Genetti et al. 2008: 123). Some languages do not require any additional affix except the nominalizer, such as in Manange, Mongsen Ao, and Zhuokeji rGyalrong (Genetti 2008: 123), with different functions. In Manange, the nominalized clause, suffixed by the nominalizer *-pɿ*, is used converbally with a causal meaning (Hildebrandt 2004:83):

<i>ŋɿ=tse mwi42</i>	<i>phrɿ42</i>	<i>kyɿ=ri52</i>	<i>pim-pɿ22</i>
(3) 1=ERG money	100	2s=LOC	give-NOM
<i>nese22</i>	<i>kyɿ52</i>	<i>kola52</i>	<i>kyu-pɿ52</i>
tomorrow	2s	clothes	buy-NOM

‘Because I gave you 100 rupees, you will buy a dress tomorrow.’

More frequently, languages combine nominalizers with additional markers, such as focus, case, or other formatives (cf. Dolakha Newar and the causal converbal suffix *-lāgin*, Genetti 2007: 463). Mongsen Ao derives its converbal formatives from a combination of nominalizers and case markers (Coupe 2007a: 443). The case markers attached to the nominalized verb indicate the function of the nominalized subordinate clause. In Meitei, for example, the locative marker attached to the nominalized verb indicates a temporal relation, the genitive indicates a purposive clause, the associative marker indicates a sequential meaning, and the ablative marker signals resultative meaning (Chelliah 1992: 242-244). In Japhug rGyalrong (Jacques 2014), the verb stem is combined with prefixes (likely nominalizers) and perfective, gerund, and purposive converbs expressing temporal succession, simultaneity, and purpose (more frequently negative purpose, since the construction commonly occurs with the negator, Jacques 2014:7), respectively.

Such a pattern of combined nominalizers and other markers is commonly found across all Tibeto-Burman (Genetti 1986, 1991, LaPolla 2006, Matisoff 1972, Herring 1991, Noonan 1997, 2005, Thurgood 1986). While it is tempting to explain the use of cases for converbal clauses in terms of semantic shifts (e.g. from agent to cause notions, from location to circumstance notions, or from dative to purposive notions), some data invite alternative explanations, for example via sentential topic constructions (cf. Bickel 1999 for Southeastern Kiranti languages; Coupe 2007a for Mongsen Ao).

The structural convergence of nominalized (and relativized) and converbal constructions is attested in many Tibeto-Burman languages. Post-head and headless nominalized relative structures are more likely to being reanalyzed as converb constructions because of their appositional function, which makes them more independent (status supported by prosodic cues, Coupe 2017b: 11). In addition, the post-head clause juxtaposes the clause-final nominalizer with other postposed markers, such as focus, determiners, postpositional clitics, which combine with the nominalizers

in the reanalysis of the converbal construction (Coupe 2007a; 2017b). An example of a semi-autonomous nominalized/relativized clause combined with a topic marker and a conditional meaning is illustrated in Mongsen Ao (Coupe 2007a: 436):

<i>ā-khū</i> (4)NRL-tiger	<i>nē</i> AGT	<i>wā</i> EXCLM	<i>səmtfà-ə̀,</i> grandchild-ANMLZ	<i>nàŋ</i> 2SG
<i>tán nē mūlūŋ-ə̀-lā</i> just AGT be.willing-ANMLZ-TOP				
<i>tàk-jùk-jù</i> weave-ASSIST-IMM		<i>kāwā</i> PTCL		

‘Tiger [replied] “Ah, grandchild, just weave with me if you are willing, alright?”’

The reanalysis of nominalized forms to converbs is common in Tibeto-Burman, from the languages of Nagaland, as seen above in Mongsen Ao, to Japhug rGyalrong (Jacques 2014) and Belhare:

<i>asamba</i> (5) last.night	<i>mai</i> person		
<i>niu-s-u-ŋ-na-ŋa</i> see-TRANS.PERF-3U-1SA-ART-ERG		<i>pɿsa</i> money	
<i>khat-lott-he</i> take-TELIC-PT			

‘The person I saw last night took the money.’
or: ‘When I saw the person, s/he took the money.’

Nominalization and relative clauses

Perhaps the most studied case of functional syncretism is between nominalization and relativization (including any noun-modifying clauses). While this phenomenon is pervasive across the Tibeto-Burman family, not all languages display this converging pattern. This overlap may surface only in some structures (cf. Rawang, LaPolla 2008), or be absent altogether, such as in Poumai Naga (Veikho 2021), among others.

Relative clauses vary with respect to order, as these structures may be pre-headed, post-headed, and internally headed³ (Bickel 1995, Noonan 2005), the latter exemplified by Belhare (Bickel 2017:715).

tombhira-ŋa wa
(6) lynx[SG]-ERGA chicken[SG.ABS]P

seiʔ-s-u=na
[3SGA-]kill-TR.PERF-3[SG]P=ART

chitt-he-m
find-PAST[-3SGP]-1PLA

'We found the chicken that the lynx had killed.'

Variable orders, pragmatically determined, are attested in many languages, such as Lahu (Matisoff 1973), Tibetan (Beyer 1992), Mongsen Ao (Coupe 2007a), Yakkha (Schackow 2015), and Chang (Coupe 2017b).

Some Kiranti languages, including Belhare, Limbu, and Athpare, feature internally headed construction types that have a fully saturated valency structure, unlike prenominal relative clauses (Bickel 1999:2), thus having the same structure as independent nominalized clauses. These structures are attested across Tibeto-Burman, and are used for various purposes, including forming questions and answers to questions, for example in Athpare (Ebert 1994; Noonan 2005) or in Bodic, where one effect is mirativity (Noonan 2005). One example of finite independent non embedded nominalization is from Lahu (Matisoff 2017: 929, glosses missing in the original).

(7) à-šwè-thâ cho-qhō nī ġā cò ve yò cê

'Once upon a time there were two thieves.'

Lit: 'It was a case of there being two thieves.'

Focus is among the core functions of independent clause nominalizations in Belhare (2017: 707, cf. example 8), Kiranti languages in general (Bickel 1999), and beyond Tibeto-Burman (Woodbury 1985).

³ Another attested structure features the correlative type, less common in Tibeto-Burman and mostly found in languages in contact with Indo-Aryan varieties (Coupe 2007b, LaPolla 2008).

- hale* *mand-u-ŋŋ-ha.*
 (8) before finish-3P-1SGA-NMZR

'It's earlier that I finished!'

3. South Asian converbs; ad sentential, peripheral subordination

Converbs, broadly defined as verb forms that function in a dependent relationship to a matrix clause, serve as a prevalent clause combining device in Tibeto-Burman and South Asian languages.⁴ The cross-linguistic literature has proposed a unifying definition of converbs that involves a constellation of properties such as non-finiteness and subordination, mainly encoding adverbial function (Haspelmath 1995; Haspelmath & König 1995). This proposed categorization does not overlap with the range of syntactic and semantic properties observed in the converbal constructions found in many languages of South Asia, for which the term South Asian converb has been proposed (Bickel 1998).

Converbs: formal variation

The South Asian converb typically involves sequences of non-final verb forms inflected with the converbal form. The final verb of the sequence may inflect with the full or partial range of verb morphology,⁵ which determines the status of the non-final verbs to the clause chain. Converb constructions can be used to join separate verbs or clauses. In the former case, the final verb is often grammaticalized, and the converbal clause forms a monoclausal syntactic unit with no intervening adjuncts or arguments between verbs (Genetti 2005). These structures have been argued to play an important role in the development of serialization (see below). Converb constructions may join separate clauses which predicates do not necessarily share arguments, and linguistic material may intervene freely. Both patterns, with shared and non-shared arguments, are illustrated in the following examples from Dolakha Newar (Genetti 2005: 41), respectively.

⁴ Also known in the Tibeto-Burman and South Asian literature as conjunctive participles and absolutive participles, reflecting its historical sources in Indo-Aryan (Davison 1981; Genetti 1988, 1994, 2005; Hale 1999; Hargreaves 1991; Jørgensen 1941; Kachru 1981; Malla 1985; Masica 1976, 1991). The converbal, non-finite marker is also termed participial. In this contribution, we use the terms South Asian converb, converbal construction and converbal form.

⁵ There are languages where the final verb of the chain can be ellipted if the information it conveys is clear from context, cf. Dolakha Newar (Genetti 2005).

<i>kehě</i>	<i>ho</i>	<i>cicā=uri</i>	<i>dāi</i>	<i>coŋ-an</i>
(9) y.sister	and	small=TOP	e.brother	stay-PART
<i>con-hin</i>				
stay-3P.PST3				

The sister and brother continued to stay (where they were).

<i>bihā</i>	<i>yeŋ-an</i>	<i>hā-en</i>	<i>ta-ene</i>	<i>āme</i>
(10) marriage	do-PART	bring-PART	put-PART	3SGEN
<i>citrānga</i>	<i>bicitrānga</i>	<i>haŋ-a</i>	<i>nis-mā</i>	<i>kae</i>
citrangga	bicitrangga	say-NR2	two-CL	son
<i>janm-ai</i>	<i>jur-a</i>			
born-BV	be-3S.PST			

‘(The king) having married and brought (her to the house), their two sons named Citrangga and Bicitrangga were born.’

The ubiquity of these structures in Tibeto-Burman allows for a wide range of structural and functional variation. Finiteness is a gradual concept in South Asian converbs (and beyond, cf. Lehmann 1988, Koptjevskaja-Tamm 1994). Many Kiranti languages use non-finite converbal constructions for modifying relationships such as manner and purpose (Bickel 1991, 1993; Ebert 2017) and employ (more) finite structures to express chaining and event sequences (Ebert 1993: 89, 2019). In Dolakha Newar and Kurtöp, for example, a converbal structure with a fully finite final verb indicates that the whole unit is sentence-final and independent. The final verb may also show non-finite morphology, such as a nominalizer, in which case the clause chain assumes the syntactic relationship indicated by it (Genetti 2005; Hyslop 2013, 2017:413). Other languages only employ non-finite structures for clause combining purposes (several Tamangic languages, most prominently Chantyal, Noonan 1999). In Tamang, the converbal construction is in free distribution with a finite construction combined with the weak verb *do*, “say”, marked by a non-finite suffix (Mazaudon 2017).

Converb functions

Converbal constructions encompass a broad range of functions in Tibeto-Burman languages and South Asia in general, one of the most prominent being narrative chaining (Bickel 1998; Coupe 2006; Genetti 2005). They serve various common

functions, including modifying relationships such as temporal overlap, manner, sequence, and purpose. Additionally, they are used for lexicalized expressions (lexical collocations, Genetti 2005), conditionals, direct quote complements (common throughout South Asia, cf. Hock 1982; Saxena 1988; Masica 1976, 1991; Noonan 1999), and recapitulation, also known as tail-head and summary-head linkage in the literature (Thompson & Longacre 1985). These structures contribute to the cohesion of narrative discourse and are found for example in Dolakha Newar (Genetti 2005) and Manange (Hildebrandt 2017). Some languages, such as Kiranti (Michailowski 2017), with some exceptions (Camling, Ebert 2003) feature specialized negation converbs denoting actions that take place without the support of the other verb.

The number of converb forms varies across languages. Some languages have a limited set of converbs, or just one converb, like the marker *-en* in Dolakha Newar (Genetti 2005) or *-(s)te* in Tibetan (DeLancey 2017:383). On the other hand, other languages have extensive converbal inventories (for example Mongsen Ao, Coupe 2007b). In languages with a more restricted inventory, the converbs tend to be less semantically loaded and more vague in meaning. The vagueness of the converbal form leads to its polyfunctionality, where the semantic relation between the converbal construction and the surrounding clauses is inferred by context, prosody or verb semantics. An example of multiple semantic interpretations is shown in Dolakha Newar, where the relationship of the last two verbs in the construction may be either sequential or modifying (Genetti 2005:43). This is illustrated below, where the verb sequence *gaen ona* can either mean ‘mount and go’ or ‘mount while riding’ (Genetti 2005:41).

<i>ām</i> (11) DEM	<i>kehẽ=uri</i> y.sister=TOP	<i>sut-pānt</i> suit-pant	<i>phi-ene</i> wear-PART	<i>tap</i> hat
<i>phi-en</i> wear-PART	<i>sara</i> horse	<i>ga-en</i> ride-PART	<i>ām</i> DEM	<i>kehẽ</i> y.sister
<i>tuj</i> EMPH	<i>on-a</i> go-3S.PST			

‘The younger sister, putting on a pants suit, putting on a hat, and riding a horse, the younger sister went.’

Conversely, in languages with a more complex converb inventory, the converbal forms tend to express individual and specific meanings. This distinction resembles the proposed distinction between ‘contextual’ and ‘specialized’ converbs (Nedjalkov 1995: 106–110), which also includes a third type, the ‘narrative’ converb. The narrative

converb, used for chaining functions, is often conflated with the contextual converb in the languages of South Asia (cf. the converbal suffix *-era* in Nepali, used for chaining and adverbial modification, Bickel 1998). In Tibeto-Burman languages, contextual, sequential, and simultaneous converbs tend to be used for clause chains (cf. for example in Tshangla, Andvik 2017), while semantically specific converbal forms “generally correlate with adverbial presupposition” (Coupe 2017:296). Sequential converbs are used to generate long text sequences in languages with extensive converb inventories, while semantically specialized converbs are typically used in a “binary structure” (Coupe 2017b:296) which consists of the non-finite converbal clause and the matrix clause. Vague, contextual converbs, as found in Dolakha Newar (Genetti 2005), allow for a wider functional range and long chain combinations but are restricted in clauses with specific and semantically explicit marking (see constraints in syndesis, Lehmann 1988).

The diachronic sources of converbs often involve nominalizers combined with case markers, which are semantically transparent in languages like Mongsen Ao and Tshangla (Coupe 2006; Andvik 2017: 294). Similar processes have been identified in Indo-European converbs. For instance, ergative, instrumental, and ablative cases are linked to converb forms with temporal and causal meanings, dative forms grammaticalize into purposive meanings, and locative evolves into conditional, concessive, and temporal meanings (Genetti 1986, Coupe 2006).

Ad-sentential and peripheral subordination

Another aspect that has been addressed in the literature is the subordinate nature of these structures (Genetti 2005). Some languages make a clear distinction between ad-sentential and peripheral subordination, following the proposals by Bickel (1991, 1993, 1998) and Van Valin (2005). These structures do not converge into the single type, of ‘adverbial subordination’ that is common in, say, Indo-European, and the main differences lie in their distributions and scope properties.

Ad-sentential subordination (and chaining) provides topic or the framework for subsequent pieces of discourse and generates sentences and text constituents, such as compound sentences, sequentialized clauses and even entire narrative paragraphs. By contrast, peripheral subordination modifies the situation provided by the main verb. Peripheral subordinated clauses are main clause constituents (“belong to its periphery”, cf. Foley & Van Valin 1984), and they are adjoined to the relational core of the clause. This distinction is formally evident in Kiranti languages, which, as seen above, use non-finite forms (converbs and supines) for peripheral subordination, while expressing ad-sentential subordination (and chaining) through (more) finite clauses (cf. Ebert 1993: 89).

Ad-sentential and peripheral clauses mainly differ in scope properties, such as illocutionary scope, as demonstrated in some Kiranti languages (Bickel 1998). In Belhare, for example, ad-sentential clauses allow illocutionary scope on either the main or subordinate clauses, but not both. In contrast, peripheral clauses attract the main clauses' scope over both main and subordinate clauses (Bickel 1993: 32):

- (12) *his-sa* *his-sa* *khar-i-ga-tlo!*
 look-SS/T look-SS/T go-2p-2-CP
 'But do go carefully!'

Ad-sentential subordinate clauses, on the other hand, are less susceptible to scope properties. This difference may be attributed to the fact that ad-sentential constituents provide frameworks, and their presence is required by global discourse structures. Their function is to keep the text coherent and intelligible, thus they are not in an immediate relationship with the main predication and are less likely to attract scope. On the contrary, peripherally subordinate clauses are main clauses constituents and elaborate on the main predication, thus more likely to attract the scope of illocutionary markers. This is because as sentential subordination are topic clauses. See for example the *-nalo* construction in Puma (Schackow et. al. 2012):

- pʌŋ pheri en-dima-bo=cha*
 (13) CONN again 1PIPOSS-female.ancestors-GEN=ADD

caha li=ku *nalo* *ahet* *mu-m*
 need be=NMLZ COND later do-1pA

'If it is necessary [to raise] our female ancestors [souls] as well, we do that afterwards.'

The *-nalo* construction functions as topic and conditional (a pattern not uncommon: Marchese 1977, Haiman 1978) and it can be further topicalized with *-na* (see above). It is outside scope and is not embedded.

4. Verb compounding and serialization

A pervasive trait of the South Asian linguistic area (Masica 1976) is the presence of ‘compound verbs’, also referred to in the literature as ‘light verbs’, ‘vector verbs’, and ‘versatile verbs’ (Matisoff 1969). These structures, extremely common throughout Tibeto-Burman, are characterized by a type of verb serialization in which the main and fully lexical verb form (V1) is followed or preceded by another verb (V2) with various degrees of lexical informativeness, ranging from grammaticalized aspectual or spatial notions to rich lexical content. The V2s are often verbs of motion, position or direction (Masica 1976). The prototypical compound verb construction is a single predicate with a composite argument structure. Similar structures are found in many Tibeto-Burman languages of Nepal, and have been studied especially in Kiranti (Ebert 1994, Bickel 1996, and Doornenbal 2009) and Kinnauri (Saxena 1992). Kiranti compound verbs differ from the Indo-Aryan counterparts in that both verbs of the compound tend to be finite to some extent. This pattern is shown in Chintang (Bickel and Zuñiga 2017: 176).

jamma
 (14) all

hod-a-thi-a-ŋs-e-ta
 break-PST[3ss]-v2:move.down.INTR-PST-PRF-IND.PST-FOC

 ‘It’s all broken down.’

The degree of finiteness of the compounded verbs, the types of verbs (transitive, intransitive, or both) involved in these constructions, and the size of the secondary verbs inventories vary across languages. In Indo-Aryan and western Kiranti languages (such as Hayu), the V1 is generally non-finite, while in eastern Kiranti the tendency is for V1 to be a partially finite form (Michailowski 2017). Kiranti compound verbs usually include both intransitive and transitive forms, sometimes subject to formal harmony or agreement in transitivity (Bickel & Zuniga 2017).

Secondary verbs inventories are more reduced in western Kiranti: in Hayu, the V2 appears restricted to the ‘motionalizer’ *la*, ‘go’ (Michailowski 2017), while Wambule Rai has a wider set of secondary verbs (Opgenort 2017). Verb of motions are among the most common sources of V2s. These forms typically encode movement away from the speaker, when translated as ‘away’, or ‘put’, indicating full completion of the event denoted by V1 (Ebert 1994). Other constructions involve stative verbs that have grammaticized as progressive markers: these are found for example in Bantawa (Boornendal 2009), Camling (Ebert 1994), and also outside Kiranti, for example in Kathmandu Newar (Hargreaves 2017), Dolakha Newar (Genetti 2005), and Suansu (Ivani 2024).

The converbal suffix found in clause chains can be at times omitted when the final verb has a grammaticalized function, and this structure often yields to verbs sequences

with no interceding material. This pattern has been argued to be an incipient case of serialization. Other conditions for serialization usually involve a converb with minimal semantic content and zero anaphora (DeLancey 1991).

Several languages show this pattern to different degrees. For example, in Kurtöp, clause chains may omit the converbal suffix when the final verb is semantically reduced (Hyslop 2013). No material can intervene between the non-final and final verbs. The interpretation of the chain is monoclausal, with the semantics provided by the lexical verb and the grammatical functions, including aspect and direction, provided by the secondary verb (Hyslop 2017: 413). A similar pattern is attested in Boro, where the converbal suffix is omitted with specific final verbs, such as *hor* ‘give’ or *lan* ‘take away’ (Boro 2012). Both structures, with and without converb, are illustrated below (Boro 2012: 8) within the same sentence (note the lack of converbal form at the end of the clause chain between the verbs *okhar* ‘come out’ and *lan* ‘take away’).

(15)	<i>i</i>	<i>p'ai-na</i>	<i>nun=k'w</i>	<i>naigir-na</i>	
	3SG	come-NF	2SG=OBJ	look.for-NF	
	<i>mun-a-ni</i>	<i>dose</i>	<i>zirai-na</i>	<i>goi</i>	
	get-NEG-BECAUSE	a.while	rest-NF	betel.nut	
	<i>k'an-se</i>	<i>za-na okhar</i>		<i>lan-bai</i>	
	CLS-one	eat-NF come.out		take-PERF	

‘He came looking for you, but not finding you, he rested, ate a betel nut, and went out.’

In Jinghpaw, verb serialization is productive: the converbal suffix is not used and verbs are under coreference and non-interruptibility constraints (Kurabe 2017: 1007). The semantics of serialization includes sequential, manner, purpose and complementation functions. In Jinghpaw, heavily grammaticized verbs may optionally occur in SVCs with no order constraints, and semantically modify the other verbs, providing meanings such as intensification, reciprocity, evidentiality and aspect (Kurabe 2017:1008). The clause chain > serialization trajectory is apparent in Tshangla, where verb concatenations lead to constructions where the semantic bleaching of the final verb is completed (i.e., *cho* < *chole* ‘to stay’, imperfective marker). In such cases, the construction denotes a unique, monoclausal event characterized by adjacency and coreference constraints (Andvik 2017: 433).

In the Tibeto-Burman languages spoken in the east, verb serialization is more systematic, although more constrained than what observed in the serializing languages of South East and East Asia (i.e., Vietnamese or Thai). In Eastern Kayah Li, the verb

complex has strict constraints in argument sharing and no intervening material is allowed between verbs. The verb forms are linked by semantic relationships of modification, sequence, and causation, the latter illustrated below (Solnit 2017: 939):

- (17) *ʔa vɪ* *jo* *cwá* *chā* *mò* *bē*
 3 throw fly go chicken female yellow
- nʌ* *rʌ*
 ASSERT PAUSE

‘She threw the yellow hen so that it flew away.’

Pwo Karen (Kato 2017: 953) features different patterns of serialized verbs, depending on the position of the negation particle. Concatenated serialization does not allow any intervening material between verbs, although their ordering is free. Their semantic relationship is action and purpose (Kato 2017:954):

- (18) *ʔəwê* *thàin* *mî*
 3SG return sleep

‘He went back and slept.’

In verb serialization with non contiguous verbs (‘separated type’, Kato 2017: 955), elements such as negation particles, NPs and adpositional phrases may intervene. Coreferentiality varies, and there are constraints on the semantics of the verbs that can occur as V2: these are restricted to volitional verbs, and the whole construction primarily has a resultative meaning, or denotes modality (Kato 2017: 955):

- (19) *jə* *ʔán* *mì* *blē*
 1SG eat rice full

‘I ate rice and got full.’

5. A questionnaire for clause combining in Tibeto-Burman

As we noted for nearly all clause combining strategies, Tibeto-Burman languages exhibit a great deal of variation while also showing striking patterns of structural convergence. In the remainder of this contribution, we propose a questionnaire to capture this variation. The questionnaire is available in the Appendix.

The structure of the questionnaire relies heavily on previous cross-linguistic categorizations of clause combining structures (especially Lehmann 1988, Cristofaro 2003, Bickel 2010, van Gijn et al. 2015). However, some of the questions have been fine-tuned to the particularities of Tibeto-Burman constructions, and additional variables have been developed to better capture this variation.

The questionnaire is designed as follows. The main unit of description (Ivani & Bickel *in press*) is a construction that corresponds to a specific clause combining structure in a given language. A single construction contains two or more (i.e. in clause chains) predication units (PUs), defined as the smallest unit of linguistic predication projecting its own syntactic environment. Each construction features individual properties captured in the questionnaire via descriptive variables. Variables are described by binary values that determine presence and form of these properties. One exception are the variables describing the position of clause linkage markers, where values correspond to specific orders (pre, post, and free) with respect to the constituent unit.

In describing constructions, we adopt a multicategorization approach (Ivani & Bickel *in press*): the individual macro domains of interest (grouped thematically in the questionnaire) are broken down into sets of variables that make up the complexity of the domain. For example, finiteness is decomposed into the uniquely identifiable variables of subject and object agreement, tense, aspect, mood, and modality (van Gijn et al. 2015). An answer can be further refined using subordinate (nested) variables. For example, differences in coding of inflectional categories can be attributed to differences in marker presence and/or differences in marker form. An excerpt of the questionnaire illustrating this in Table 1:

ID	Question	Values
1	Are there differences in how subject agreement is coded on the PUs?	Y/N/NA
1.1	Is the difference due to the presence of marking?	Y/N
1.2	Is the difference due to the form of marking?	Y/N
2	Are there differences in how object agreement is coded on the PUs?	Y/N/NA
2.1	Is the difference due to the presence of marking?	Y/N
2.2	Is the difference due to the form of marking?	Y/N
3	Are there differences in how tense is coded on the PUs?	Y/N/NA
3.1	Is the difference due to the presence of marking?	Y/N
3.2	Is the difference due to the form of marking?	Y/N

Table 1: Excerpt from the clause combining questionnaire.

Questions 1-7 capture the common finiteness asymmetries found in clause combining structures. Questions 8-12 are dedicated to nominalization in terms of the presence (or absence) of a nominalizer marker (cf. §2), and capture noun-like properties (van Gijn et al. 2015) of the PUs, including the presence of case, number, and gender markers.

Questions 12-12.4 deal with coreference constraints across PUs, as these are found in many clause combining structures (cf. §2 and §4). Question 13 describes whether the relationship between the PUs is flagged (i.e., clausal nominalization can be encoded via an overt marker or by juxtaposition, such as in Karbi, cf. §2). Questions 14-15 cover clause chains properties. Questions (16-20) describe syntactic functions of PUs constituents (as for example the functions described in §2 and §3) and related constraints in addition to extraction possibilities (cf. Bickel 2010) from clauses.

Questions 21-25 describe clause attachment properties (including linearization). Attachment properties refer to the observed distinction between ad-sentential and ad-peripheral subordination (cf. §3; see also Bickel 1998 and 2010). These questions also address the linearization of the PUs and the integration between the PUs with respect to adjacency constraints. Questions 26-34 describe scope constraints of illocutionary force, source, focus, and negation, following the typology proposed in Bickel 2010. Questions 35-37 deal with verb selection constraints and lexical vs. non-lexical verb usage (cf. §4).

The remaining questions (38-52) describe the semantic relation types of the constructions; these variables rely heavily on the categorization proposed by Cristofaro (2003) in her typology of subordination (cf. van Gijn 2014 and van Gijn et al. 2015 for its application to areal and family-specific studies, respectively).

Each construction is defined as a unique set of entries in the questionnaire: two constructions differ if there is a difference in at least one variable. This approach, which has been proven fruitful in describing other linguistic domains (cf., for example, the NP structure domain in AUTOTYP, Bickel et al. 2023; cf. also Witzlack et al. 2022), comes with several practical advantages, primarily flexibility. Compilers can expand the questionnaire with additional variables that reflect fine-grained, language-specific differences across clause-combining constructions.

This questionnaire is thus a ‘starter kit’ for compilers and language experts: while it offers a ready, standalone tool able to capture systematically a wide range of properties in various clause combining types, from converbal clauses to clause chains and serialization, it can be expanded to describe particulars in clause linkage without ‘reworking’ the rest of the questionnaire. The questions are designed to be self-contained, which allows a straightforward extension of the questionnaire. Another apparent advantage of this structural flexibility is that the final description for each construction is closer to the variation we observe in the concrete data. By extending the questionnaire, we describe and capture variation rather than compress or discard it,

revealing a picture of systematized complexity and diversity that is informative to typology and closer to language-specific analyses.

6. Summary

In this contribution, we reviewed some of the most prominent clause combining structures observed in Tibeto-Burman languages: nominalization, converbal clauses, and compound clauses. We observe a rich, complex picture of structural and functional variation and convergence. In the second part of the chapter, we propose a ‘starting kit’ questionnaire to capture, disentangle and systematize this diversity. The goal is ultimately to offer a flexible and expansible tool that describes variation rather than discarding it, bridging typology and language-specific descriptions.

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Appendix: Questionnaire

ID	Question	Values	Explanation
1	Are there differences in how subject agreement is coded on the PUs?	Y/N/NA	These questions pertain to the inflectional possibilities featured in the predicated units. The categories individually represented feature subject and object agreement, tense, aspect, mood, and epistemic/evidential categories. See Bickel 2010 (finiteness), Cristofaro 2003, van Gijn et al. 2015 for discussion.
1.1	Is the difference due to the presence of marking?	Y/N/NA	
1.2	Is the difference due to the form of marking?	Y/N/NA	
2	Are there differences in how object agreement is coded on the PUs?	Y/N/NA	
2.1	Is the difference due to the presence of marking?	Y/N/NA	
2.2	Is the difference due to the form of marking?	Y/N/NA	If the language does not code the respective inflectional category, the question should be answered with 'NA' (not applicable). The sub-questions Q.x are only relevant if the main question Q is answered with 'Yes' (Y), otherwise they are to be answered with 'NA' (not applicable).
3	Are there differences in how tense is coded on the PUs?	Y/N/NA	
3.1	Is the difference due to the presence of marking?	Y/N/NA	
3.2	Is the difference due to the form of marking?	Y/N/NA	
4	Are there differences in how aspect is coded on the PUs?	Y/N/NA	
4.1	Is the difference due to the presence of marking?	Y/N/NA	
4.2	Is the difference due to the form of marking?	Y/N/NA	
5	Are there differences in how mood is coded on the PUs?	Y/N/NA	

5.1	Is the difference due to the presence of marking?	Y/N/NA	
5.2	Is the difference due to the form of marking?	Y/N/NA	
6	Are there differences in how epistemic categories are coded on the PUs?	Y/N/NA	
6.1	Is the difference due to the presence of marking?	Y/N/NA	
6.2	Is the difference due to the form of marking?	Y/N/NA	
7	Are there differences in how evidential categories are coded on the PUs?	Y/N/NA	
7.1	Is the difference due to the presence of marking?	Y/N/NA	
7.2	Is the difference due to the form of marking?	Y/N/NA	
8	Is a PU marked with a nominalizer?	Y/N	
9	Is a PU marked with a case marker?	Y/N	
10	Is a PU marked with a number marker?	Y/N	
11	Is a PU marked with a gender marker?	Y/N	
12	Are there coreference constraints between the PUs?	Y/N	These questions address coreferentiality constraints across PUs, as these are found in many clause combining structures in T-B and beyond (cf. subsection <i>Clausal nominalization and complementation</i>).
12.1	Does the A argument of at least one of the PUs have to be coreferent with an argument of some other PUs?	Y/N/NA	
12.2	Does the S argument of at least one of the PUs have to be coreferent with an argument of some other PUs?	Y/N/NA	

These questions aim at determining the noun-like properties of the PUs. Variables include case, gender, and number marking, along the categorization proposed in van Gijn 2014 and van Gijn et al. 2015.

These questions address coreferentiality constraints across PUs, as these are found in many clause combining structures in T-B and beyond (cf. subsection *Clausal nominalization and complementation*).

The sub-questions 12.x are only relevant if the main question 12 is answered with 'yes' (Y), otherwise they are to be answered with 'NA' (not applicable).

12.3	Does the P argument of at least one of the PUs have to be coreferent with an argument of some other PUs?	Y/N/NA	Every single PU shares the same reference, for example in a long clause chain structure (cf. example (15) from Boro in section <i>Verb compounding and serialization</i>).
12.4	Do all the the PUs have to share one reference?	Y/N	
13	Does at least one PU use a dedicated linkage marker?	Y/N	
13.1		pre/post/free/NA	
	What is the location of the linkage marker in relation to its constituent PU?		
14	Can there be more than two PUs within the construction?	Y/N	The sub-question 13.1 is only relevant if the main question 13 is answered with 'yes' (Y), otherwise it is to be answered with 'NA' (not applicable).
15	Do the PUs form a clause chain?	Y/N	
16	Is a PU coded as a verb complement?	Y/N	
17	Is a PU coded as a noun complement?	Y/N	These questions address and flag clause chains.
18		Y/N	
	Is a PU coded as a converbal clause?		
19	Is a PU a relative clause?	Y/N	This non-exhaustive list of questions aims at capturing the range of syntactic functions in T-B clause combining structures, i.e. nominalization and complementation (cf. <i>Nominalization and complement clauses</i> section). These questions also cover more complex structures, such as double complementation embedding, found for example in Mongsen Ao, cf. <i>Nominalization</i> Section)
19.1	Can a relative clause PU occur before its head?	Y/N/NA	
19.2	Can a relative clause PU occur after its head?	Y/N/NA	
19.3	Can a relative clause PU be embedded inside its head?	Y/N/NA	

The sub-questions 19.x are only relevant if the main question 19 is answered with 'yes' (Y), otherwise they are to be answered with 'NA' (not applicable).

20	Are there constraints regarding extraction from PUs?	Y/N	This variable addresses extraction of elements of the PUs, following the categorization proposed in Bickel 2010.
21	Is a PU adjoined ad-sententially?	Y/N	These questions capture the layer of attachment of adjoined clauses, following the categorization proposed by Bickel 2010. Cf. <i>Ad-sentential and ad-peripheral subordination</i> section. An example of ad-peripheral attached clause is illustrated by Belhare in (12) and of ad-sentential subordination is illustrated in Puma in (13).
22	Is a PU adjoined ad-peripherally?	Y/N	
23	Is a PU detached?	Y/N	
24	Are at least two PUs obligatorily juxtaposed?	Y/N	These questions pertain to the linear properties of the PUs and the integration between the PUs with respect to adjacency constraints.
25	Are all PUs obligatorily juxtaposed?	Y/N	
26	Can the illocutionary force operator apply to multiple PUs (with the rest being presupposed)?	Y/N	These questions address the scope of illocutionary force operators, along the categorization proposed in Bickel 2010.
27	Can the illocutionary force operator apply to at most one PU (with the rest being presupposed)?	Y/N	
28	Is a PU prohibited from receiving illocutionary force (has to be presupposed)?	Y/N	
29	Is there one source/operator of tense that affects multiple PUs?	Y/N	These questions address the scope of source operators, along the categorization proposed in Bickel 2010.
30	Can there be multiple independent sources/operators of tense over the PUs?	Y/N	
31	Can negation apply to multiple PUs?	Y/N	These questions address the scope of negation, along the categorization proposed in Bickel 2010.
32	Can there be multiple independent negators?	Y/N	
32.1	Are there differences in how the independent negators are coded?	Y/N/NA	The sub-question 32.1 is only relevant if the main question 32 is answered with 'yes' (Y), otherwise it is to be answered with 'NA' (not applicable).

33	Does focus simultaneously apply over multiple PUs?	Y/N	These questions address the scope of focus operators, along the categorization proposed in Bickel 2010.
34	Can multiple PUs be independently focused?	Y/N	
34.1	Are there differences in how the focus is coded on different PUs?	Y/N/NA	The sub-question 34.1 is only relevant if the main question 34 is answered with 'yes' (Y), otherwise it is to be answered with 'NA' (not applicable).
35	Is a PU restricted to a specific set of verbs?	Y/N	This question addresses whether a PU is restricted to a limited set of verbs. See for example the reduced inventories of secondary verbs typical of western Kiranti languages (cf. Section <i>Verb compounding and serialization</i>).
36	Is a PU a lexical verb that is semantically reduced within the construction?	Y/N	
37	Is a PU a non-lexical verb?	Y/N	These questions address whether the construction features semantically reduced verbs with a grammaticalized meaning. Cf. <i>Verb compounding and serialization</i> section.
38	Can this construction express successive temporal relations?	Y/N	
39	Can this construction express simultaneous temporal relations?	Y/N	These questions address semantic relation types across PUs, based on the categories proposed in Cristofaro 2003 (cf. also van Gijn 2014, van Gijn et al. 2015).
40	Can this construction express causal relations?	Y/N	
41	Can this construction express locational relations?	Y/N	
42	Can this construction express purpose relations?	Y/N	
43	Can this construction express condition relations?	Y/N	
44	Can this construction express counterfactual conditional relations?	Y/N	
45	Can this construction express	Y/N	

	concessive relations?		
46	Can this construction express desiderative relations?	Y/N	
47	Can this construction express cognition relations?	Y/N	
48	Can this construction express perception relations?	Y/N	
49	Can this construction express indirect utterance relations?	Y/N	
50	Can this construction express resultative relations?	Y/N	
51	Can this construction express manner relations?	Y/N	
52	Can this construction express motion relations?	Y/N	